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RFID CARD READER

UNDERGRADUATE PROJECT
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1 Abstract

This undergraduate project was conducted in a laboratory course called “*Electronic Circuits: Laboratory course*” in team with my classmate, Alexandros Symeonidis. In this project, a detection system was developed using an Arduino UNO, an RFID module [Radio Frequency Identification], specifically the RC522, and some LEDs and buzzer for the user interface. The interconnections between the RFID module, the LEDs and the buzzer were implemented on a prototyping board, while those of the Arduino were implemented with the use of jumper wires and pins that were soldered on the prototyping board.

RFID technology has a really broad spectrum of applications, from banking operations to everyday transport. The purpose of the read-out system that we developed was to scan RFID tags and evaluate whether the owner is allowed to access an area such as a gym, a swimming pool, and the like, or not.

2 RFID technology

RFID is a wireless communication technology that is used to identify objects within a specified radius. An RFID system consists of two main components. The first is the transceiver, also known as RFID tag, which is a small chip that consists of an integrated circuit, including memory for storing information, and an antenna. The second part is the reader or sensor, which reads the data from the RFID tags. RFID readers have an integrated antenna and a control unit. A basic RFID system is depicted in figure (1.1). The tag is activated by the electromagnetic wave emitted

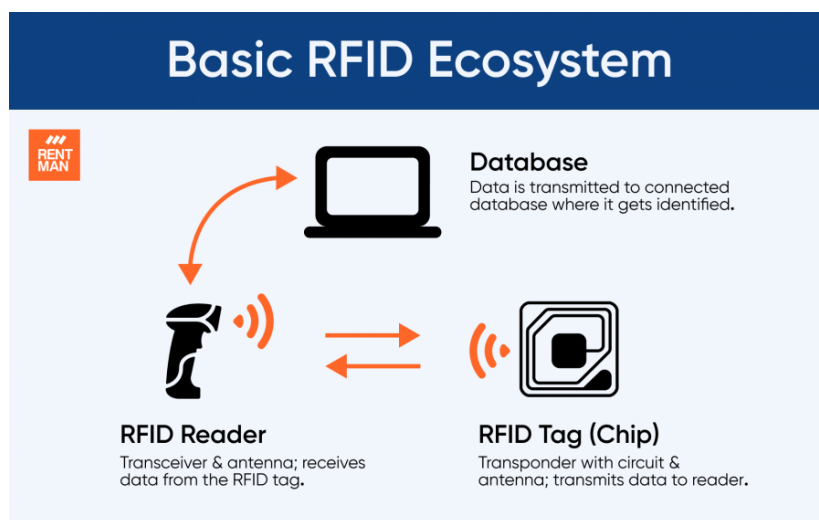


Figure 1.1: Basic RFID system.

from the RFID reader, and afterwards it transmits an identification number back to the reader. The identification number is then used for the inventory of tagged items (Card ID).

Furthermore, tags can be passive or active. Passive tags are powered only by the incoming electromagnetic wave from the reader and therefore have a shorter operating range. Active tags are powered by a battery and can have a longer range, up to hundreds of meters. This project was implemented using a passive RFID tag.

3 Pinout

As mentioned above, the main components of the RFID detection system were the Arduino UNO board and the RFID module (RC522). The communication protocol used was SPI, in which four connections are needed for the Arduino board and the RC522 to communicate effectively. Specifically, the wires are the following:

- SCLK: The wire for the clock signal.
- MOSI (Master Output Slave Input): The wire responsible for data transmission from the master device to the slave device.
- MISO (Master Input Slave Output): The wire responsible for data transmission from the slave device to the master device.
- CS (Chip Select): The wire responsible for the selection of the slave device with which the microcontroller is about to communicate.

The great advantage of this type of communication over the I²C is that the data transmission is duplex. In particular, a device can send data while receiving data.

For the operation of RC522, a ground and voltage supply wire, as well as a reset wire, were also necessary. The role of the reset pin is that when its value is low, internal current sinks including the oscillator are turned off and the input pins are disconnected from the outside world. On the rising edge, the module is reset. It is worth mentioning that an interrupt pin is also present in the RC522 module, in order to alert the microcontroller when an RFID tag comes into its vicinity. However, in this case, it remained unutilized. In the following tables, the connections between the Arduino UNO and the RC522 and those between the Arduino UNO and the UI devices are presented, respectively. In addition, in figures (1.2) and (1.3) the schematic and the layout of the RFID system are depicted, respectively.

Table 1: Connections between the Arduino UNO and the RC522 module.

Signal	RC522 pins	Arduino UNO pins
RST/Reset	RST	9
SPI SDA	SDA	10
SPI MOSI	MOSI	ICSP-4
SPI MISO	MISO	ICSP-1
SPI SCK	SCK	ICSP-3

Table 2: Connections between the Arduino UNO and the UI devices.

UI devices	Arduino UNO pins
Buzzer	4
Greek LED	5
Red LED	6
Blue LED	7

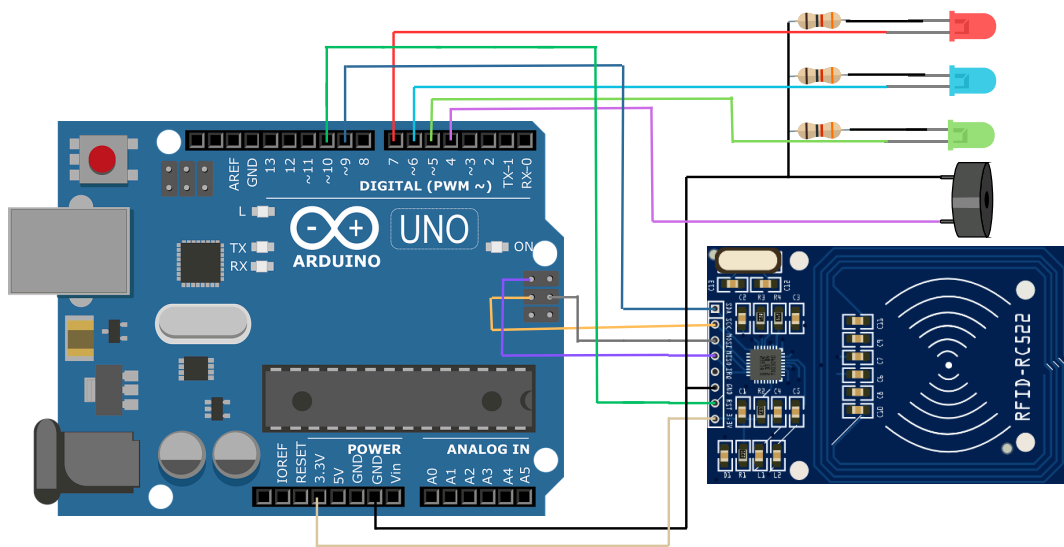


Figure 1.2: Schematic design.

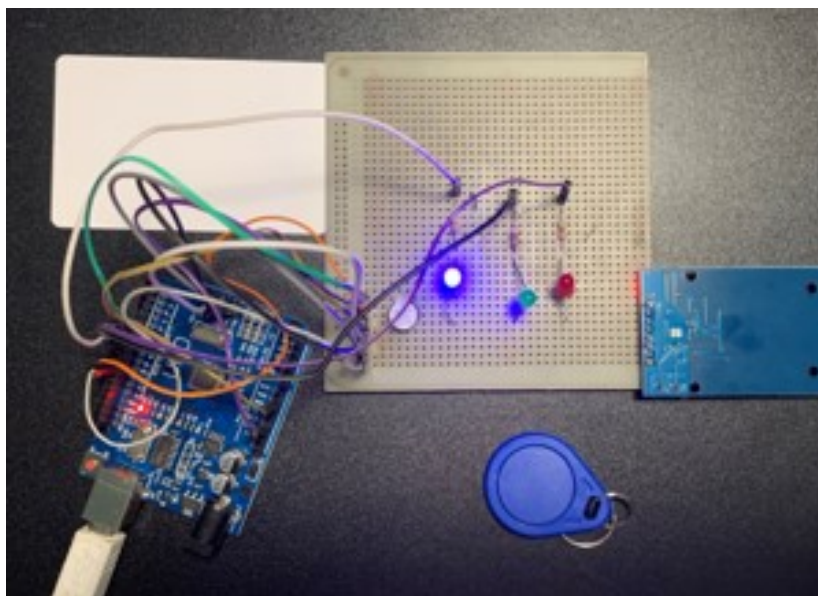


Figure 1.3: Layout design.

4 Tag's Memory

Before generating the detection code, our names were written in the tag's memory in order to be illustrated when the tag is detected by the RC522 module. For this process, we used the "rfid_write_personal_data.ino" file from a GitHub's repository. Each RFID tag contained one name.

Regarding the RFID tag's memory, it consisted of 16 sectors of four 16 bytes blocks, as can be clearly seen in figure (1.4). The total memory capacitance was 1024 bytes, of which some blocks contained information and thus there were certain blocks that are available for data storage. In particular, the block[3] of each sector contains information that grants read and write access to the remaining blocks in a sector, and it is called sector trailer. Furthermore, the Block[0] of Sector[0] is called manufacturer block, and it contains IC manufacturer data, and the Unique Identifier (UID).

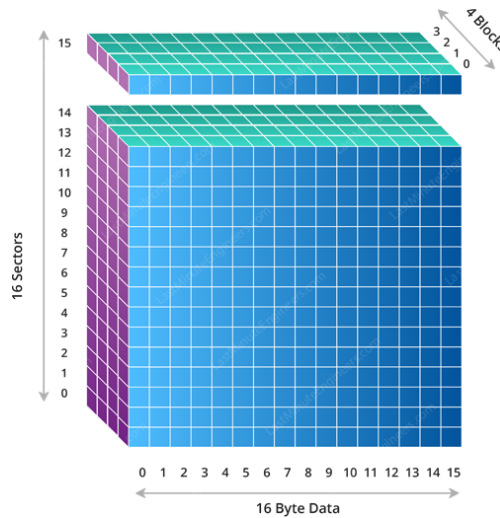


Figure 1.4: RFID tag's memory.

As a consequence, we chose to write our first name in Sector[0], Block[2] and our last name in Sector[0], Block[1] of each RFID tag.

5 Code

The code was generated in order to identify three stages of the RFID system. The first is when no RFID tags are present in front of the RFID sensor, for which the blue LED is activated. In the second stage, the green LED is activated when the tag that is detected by the sensor contains a valid ID. In the third stage, the red LED and the buzzer are activated when the tag that is detected by the sensor contains an invalid ID.

After writing the data in the RFID tags, the detection code was generated which was based in `rfid_read_personal_data.ino` and `DumpInfo.ino` files from the previous repository. The code in the `DumpInfo.ino` file contains functions that are meant to identify the card ID of each tag, which is unique for each tag (the card ID is a string of 4 bytes, written in the hexadecimal numbering system). The card ID [43, E7, B3, A6] was considered valid and the green LED would be activated whenever the RC522 module detected this particular tag. Regarding the other tag, the red LED and the buzzer would be activated whenever it was detected.

In this part of the report, the RFID card reader code is examined. The code is also uploaded to my personal GitHub repository. For the first 17 lines of code, it becomes apparent that the necessary libraries are included and the pins, `RST_PIN` (Reset) and `SS_PIN` (Chip select), are defined. Moreover, the object of the RC522 and the struct, which will hold the card information (it is used in the `readBlock` function that is presented in the code, but its analysis is not discussed in this report), are being created. Lastly, a table containing the valid card ID and a table that is going to host the content of a block are generated.

```
1 #include <SPI.h>           //SPI bus library
2 #include <MFRC522.h>       //RC522 library
3
4 #define RST_PIN            9           // reset pin
5 #define SS_PIN             10          // slave select pin
```

```

6
7 MFRC522 mfrc522(SS_PIN, RST_PIN); // Create MFRC522 object
8 MFRC522::MIFARE_Key key;           //create a MIFARE_Key
      struct named 'key', which will hold the card information
9
10 byte accessCard[4] = {0x43, 0xE7, 0xB3, 0xA6};
11 byte readtheblock[16];
12 int i;
13 int counter = 0;
14 int greenPin = 5;
15 int redPin = 6;
16 int bluePin = 7;
17 int buzzerPin = 4;

```

Listing 1: Libraries that were used.

In the setup code, every byte of the key gets the value 255, and the SPI protocol as well as the RC522 module are being initialized.

```

1 void setup() {
2   pinMode(greenPin, OUTPUT);
3   pinMode(redPin, OUTPUT);
4   pinMode(bluePin, OUTPUT);
5   pinMode(buzzerPin, OUTPUT);
6   Serial.begin(9600); // Initialize serial communications
      with the PC
7   SPI.begin(); // Init SPI bus
8   mfrc522.PCD_Init(); // Init MFRC522
9   Serial.println(F(" Scan your card..."));
10  for (byte i = 0; i < 6; i++)
11    key.keyByte[i] = 0xFF; //Security key, bytes are 255

```

12 }

Listing 2: Setup code.

In the first 15 lines of the loop code, the blue LED is activated (first stage) indicating that an RFID tag is awaited to pass. If an RFID tag is present, the first if statement is not executed and if the RFID tag can be read the second if statement is not executed as well. Since the identification of a tag has been successful, the blue LED is deactivated and a message appears on the screen.

```
1 void loop() {
2
3   digitalWrite(bluePin, HIGH);
4
5   // Reset the loop if no new card present on the sensor/
   // reader. This saves the entire process when idle.
6   if ( ! mfrc522.PICC_IsNewCardPresent() ) {
7       return;
8   }
9
10  // Select one of the cards
11  if ( ! mfrc522.PICC_ReadCardSerial() ) {
12      return;
13  }
14  digitalWrite(bluePin, LOW);
15  Serial.println("your card has passed successfully");
16
```

Listing 3: Loop code (first stage).

In the next stage, the block [2] that contains the name of the RFID tag is read and the validation of the RFID tag is examined. To be more specific, the four bytes that define the card ID are compared with the valid ID that we initialized at the beginning. If

the counter is equal to 4, that means that the RFID tag is indeed valid and therefore a series of prints are executed, and the green LED is activated (second stage). If the counter has obtained a different value, the RFID tag permission is declined, and thus the red LED as well as the buzzer are activated, and a series of prints are executed (third stage). The role of the function `mfr522.PCD_StopCrypto1` is to allow the communication with other RFID tags after a valid one has passed.

```
1  readBlock(2, readtheblock);
2  for (i=0; i<4; i++)
3  {
4      if (mfr522.uid.uidByte[i] == accessCard[i])
5          counter++;
6  }
7  if (counter == 4){
8      digitalWrite(greenPin, HIGH);
9      Serial.print("Hello");
10     for (int j=0 ; j<16 ; j++)
11     {
12         Serial.write (readtheblock[j]);
13     }
14     Serial.println("");
15     Serial.println("WELCOME!!!");
16     counter = 0;
17     delay(2000);
18     digitalWrite(greenPin, LOW);
19 }
20 else{
21     digitalWrite(redPin, HIGH);
22     digitalWrite(buzzerPin, HIGH);
23     Serial.print("Oops");
```

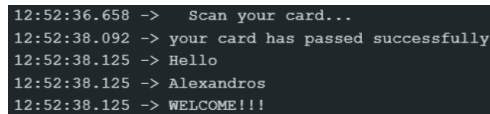
```
24     for (int j=0 ; j<16 ; j++)
25     {
26         Serial.write (readtheblock[j]);
27     }
28     Serial.println("");
29     Serial.println("you are not in the list, sorry...");
30     counter = 0;
31     delay(2000);
32     digitalWrite(redPin, LOW);
33     digitalWrite(buzzerPin, LOW);
34 }
35 mfrc522.PCD_StopCrypto1();    //for the RC522 to
    communicate with PICCs after communicating with an
    authenticated PICC
36 Serial.println(F("  Scan your card...")); //Message to scan
    the card
37 }
```

Listing 4: Loop code (second and third stages).

6 Operation & Results

The steps of the operation are:

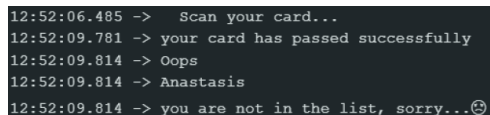
1. We upload the code that we generated in the Arduino UNO from our PC, and afterwards we unplug the RFID system from our PC, powering the RFID system from an external source (e.g: battery).
2. When the blue LED is activated, the RFID system is ready for use. If we had an LCD monitor connected to the Arduino, the message "Scan your card..." would be illustrated.
3. Depending on which tag (keychain or card) is detected by the RC522 module, the following operations are expected to occur:
 - (a) Keychain tag: The green LED is activated, and if we had an LCD monitor connected to the Arduino, the following message would be illustrated.



```
12:52:36.658 -> Scan your card...
12:52:38.092 -> your card has passed successfully
12:52:38.125 -> Hello
12:52:38.125 -> Alexandros
12:52:38.125 -> WELCOME!!!
```

Figure 1.5: Serial monitor message for the keychain tag detection.

- (b) Card tag: Both the red LED and the buzzer are activated, and if we had an LCD monitor connected to the Arduino, the following message would be illustrated.



```
12:52:06.485 -> Scan your card...
12:52:09.781 -> your card has passed successfully
12:52:09.814 -> Oops
12:52:09.814 -> Anastasis
12:52:09.814 -> you are not in the list, sorry...☹
```

Figure 1.6: Serial monitor message for the card tag detection.